

8 The role of red blood cells is to transport oxygen around the body.

The oxygen-carrying capacity of red blood cells varies from species to species.

The oxygen-carrying capacity depends partly on the size and structure of the red blood cells and their haemoglobin content.

(a) Describe the structure of a haemoglobin molecule.

(3)

*(b) The table shows some information about red blood cells from four species of air-breathing mammals.

The species are listed by decreasing size.

Species of mammal	Mean number of red blood cells per dm ³ of blood	Mean mass of haemoglobin per cell / pg	Mean cell volume / μm ³	Oxygen-carrying capacity / cm ³ oxygen per dm ³ of blood
Beluga whale	3.3×10^{12}	57	134	259
Camel	1.1×10^{13}	14	319	211
Man	5.0×10^{12}	29	90	196
Goat	1.6×10^{13}	7	18	139



Explain the relationships between the sizes of these species of mammal, the characteristics of their red blood cells and their oxygen-carrying capacity.

Use the information in the table to support your answer.

(6)

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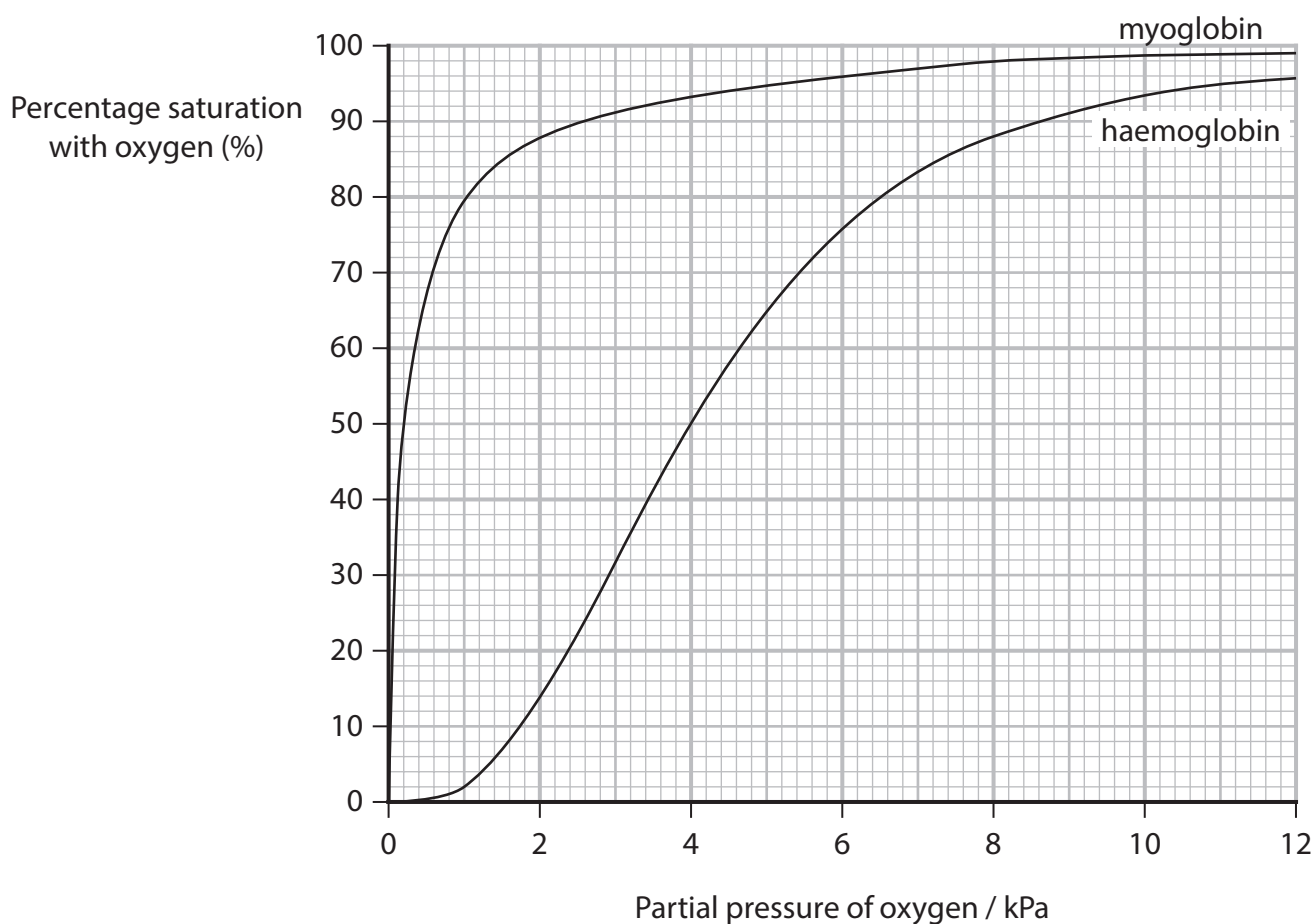
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- (c) The graph shows the oxygen dissociation curve for haemoglobin and myoglobin. Myoglobin is present inside muscle cells and acts as an oxygen store.



- (i) The decrease in percentage saturation with oxygen of myoglobin from 6 kPa to 4 kPa is 3%.

Calculate the difference between this decrease and the decrease for haemoglobin from 6 kPa to 4 kPa.

(2)

Answer%



- (ii) Describe the effect of the partial pressure of oxygen on the percentage saturation with oxygen of myoglobin.

Use values from the graph to support your answer.

(2)

- (iii) Suggest why the dissociation curve for myoglobin is to the left of the curve for haemoglobin.

(2)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

